Daily UTC delivery lateness CSV report

The system in one sentence

Join two daily files and list orders delivered more than one day late.

Key design calls

Use code for the join. Rejected: A model. Why: Dates need exact rules

Save one complete report. Rejected: Partial files. Why: Staff need clear proof

Keep the tool local. Rejected: Remote writes. Why: The brief needs only files

Who acts and who decides

Action	Autonomy
Check and join the files	acts alone
Save the local report	acts alone
Resolve bad input files	human decides

Top risks by RPN

FM3 | Incomplete, malformed or stale evidence | RPN 128

FM2 | Threshold or UTC-window error | RPN 96

FM4 | Partial or false-complete local publication | RPN 96

FM8 | Failure is not owned or seen | RPN 96

FM9 | Exposure or uncontrolled retention | RPN 72

Launch blockers and owners

H1, H2, H4, H7: The proposed CSV contract is...; Operations supplies two complete snapshots...; Accepted timestamps use YYYY-MM-DDTHH:MM:SS[.ffffff]Z or...; Keep committed CSVs, manifests and... Owner: Operations data owner.

H3, H5, H8: The daily report covers actual...; Proposed operational budget is at...; A date with an existing... Owner: Operations report owner.

H6: Run on one trusted local... Owner: Implementation owner.

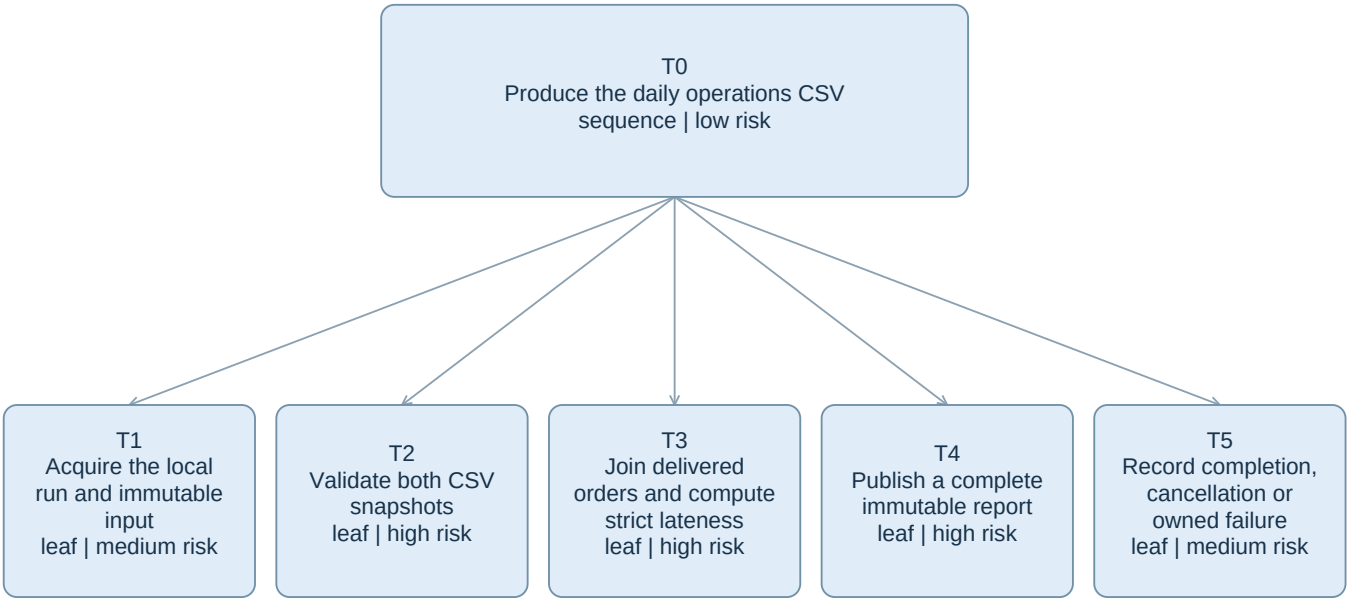
Illustrative cost per case

USD 0.00000 per case in the model budget. This is an estimate, not a current provider quote. Staff and tools cost extra.

Review status: self-review; round 3; pass.

Proposed design. Full rules and blockers: design.json and assumptions.md.

Task tree



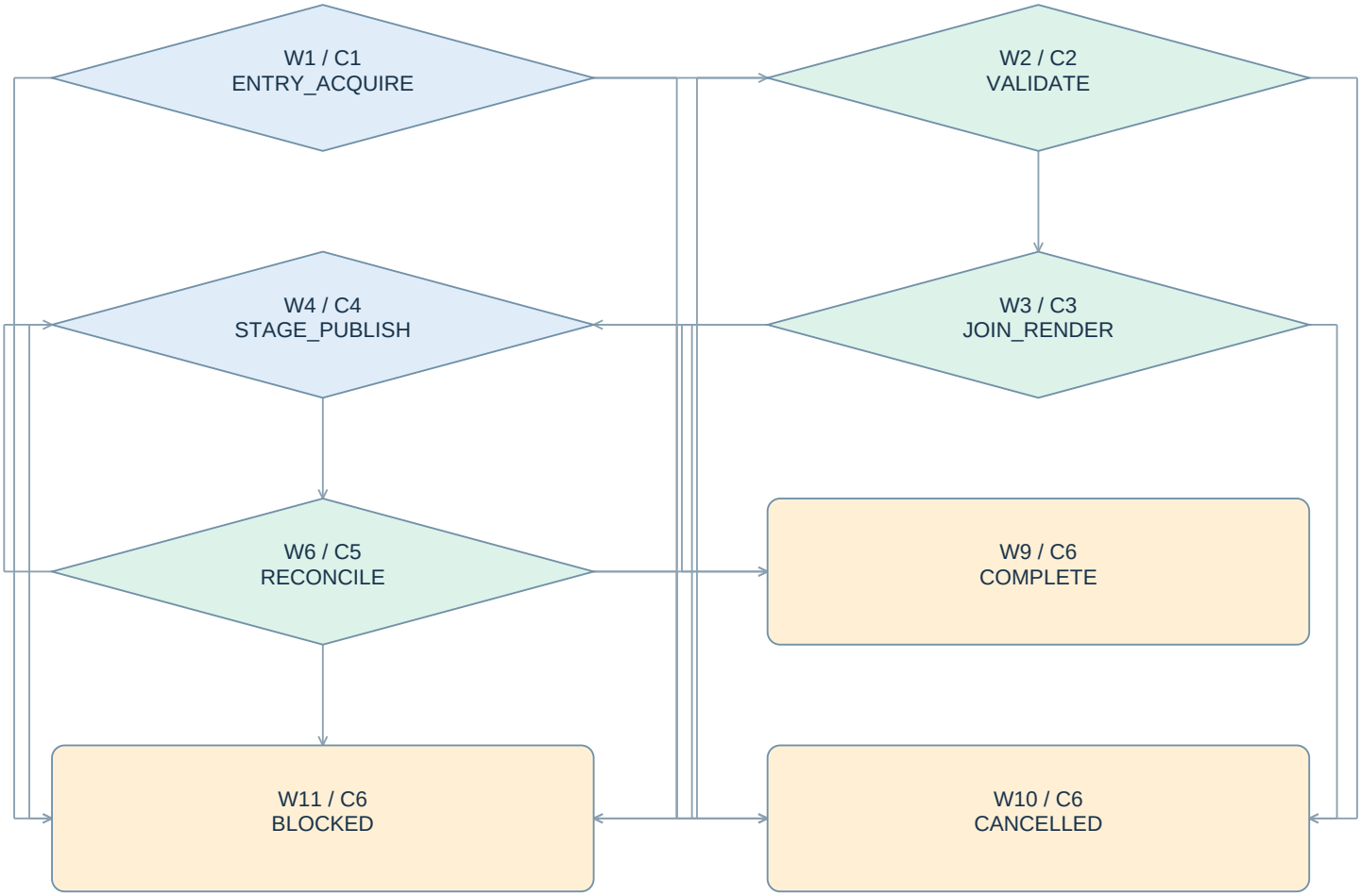
Labels show each leaf mode and risk. Full task inputs, examples and limits are in design.json and PRD.md.

Plain words and scope

The tool writes a local CSV. It does not send messages, forecast delays or change any order.

A fingerprint is a check that the approved details have not changed. A record ID is proof from a tool that the change was saved.

Workflow diagram



Diamonds mark choices. Arrows show all routes, including retry and recovery. Exact branch rules are in architecture.md.

Colour	Role
#E8E0FA	Agents (LLM)
#E0EDF8	Code: tools
#DDF2E9	Code: validation
#FFF0D5	Code: human triggers

Top risks and release checks

Risk heat table - severity x occurrence

Each cell names the risks at that severity and occurrence. The full scores, causes, controls and owners are in fmea-register.csv.

S / O	1	2	3	4	5	6	7	8	9	10
10	-	-	-	-	-	-	-	-	-	-
9	-	-	-	-	-	-	-	-	-	-
8	-	FM6	FM2, FM4	FM3	-	-	-	-	-	-
7	-	-	FM1, FM5, FM7	-	-	-	-	-	-	-
6	-	-	FM9	FM8	-	-	-	-	-	-
5	-	-	-	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-	-	-	-
3	-	-	-	-	-	-	-	-	-	-
2	-	-	-	-	-	-	-	-	-	-
1	-	-	-	-	-	-	-	-	-	-

Risk	Control summary / full register linked
FM1 Wrong or duplicate order identity	Exact case-sensitive H1 ID validation, fail duplicate IDs; same-namespace source procedure H2.
FM2 Threshold or UTC-window error	Parse UTC exact microseconds, strict >, half-open UTC day and sorted deterministic output.
FM3 Incomplete, malformed or stale evidence	Whole-file validation and immutable snapshot hash; fail missing delivered promise; H2 source cutoff procedure. No inferred promise or silent skip.
FM4 Partial or false-complete local publication	Stage and fsync before atomic directory publication; manifest+hash required; read-only reconciliation before one bounded retry; unknown stays blocked.
FM5 Concurrent or conflicting re-publication	Per-date exclusive lock; same input+config no-op; changed existing date conflicts; never auto-break lock or auto-replace.
FM6 Unsafe input or filesystem scope	H1 safe ID profile; parse as inert CSV with no shell/eval; allowlisted real paths and no symlinks; fresh local permission check...
FM7 Cancellation, deadline or size-limit mishandling	Byte/row/time bounds, cancellation before rename and deferred outcome reconciliation after rename begins; no blind retry.
FM8 Failure is not owned or seen	Exit code + local durable status, stderr fallback, daily inspection and manual acknowledgement/escalation in H5; never send email or remote updates.
FM9 Exposure or uncontrolled retention	Persist minimized columns/hashes/counts; error row numbers only; dedicated private local root; H7 approval and manual retention procedure.

Risk scores are proposed estimates. They are not measured odds or risk acceptance.

Components and contracts

This inventory is a summary. Exact typed inputs, outputs and permission rules are in architecture.md and the build milestone files.

Component / group	Responsibility	Task IDs
C1 / Code: tools	Run controller and immutable read	T1
C2 / Code: validation	CSV integrity validator	T2
C3 / Code: validation	Exact join and renderer	T3
C4 / Code: tools	Atomic local publisher	T4
C5 / Code: validation	Read-only publication reconciler	T4
C6 / Code: human triggers	Local outcome recorder	T5

Shared control boundary

Code checks identity, money, dates and limits. A model may suggest a choice. It cannot authorise or perform a business write.

Architecture and model budget

F1 - Code workflow fit

Choice: A deterministic Python CLI is the complete first release; zero agent framework and zero model calls.

Rejected: Rejected LLM/multi-agent system: adds parsing uncertainty, credentials, cost and orchestration with no judgement task.

Trade-off: Both options assessed on exact join/date correctness, local deployment, retries/audit and operating burden; code directly satisfies every criterion. No vendor capability claim...

F2 - One-host durable batch architecture

Choice: C1→C2→C3→C4 with C5 reconciliation and C6 local terminal status. Per-date exclusive lock, content-bound run identity and immutable report directory.

Rejected: Rejected database/workflow service for this small local report: extra infrastructure and operational state without a remote or long-running approval requirement.

Trade-off: Local filesystem behavior is a launch dependency; a future distributed execution would require a new concurrency/publication design.

F3 - Schedule and bounded orchestration

Choice: Invoke CLI manually first; use an existing local daily scheduler at proposed 08:00 UTC after export availability is verified. One run bounded to 120s plus 5s...

Rejected: Rejected cron-like remote service provisioning: outside design scope and not necessary for a small local job.

Trade-off: Manual invocation and an existing scheduler have the same validation, locks, exact data and publication rules; scheduling only changes trigger ownership. No...

F4 - Filesystem authority and inert inputs

Choice: Treat CSV content as untrusted data; strict parser/ID profile, no eval or shell interpolation. Resolve paths against allowlisted roots, forbid symlinks, fresh capability checks and exclusive...

Rejected: Rejected processing arbitrary paths and preserving formula-like IDs directly in spreadsheet CSV: unclear scope and formula risk.

Trade-off: The narrow ID profile is a confirmable source-contract dependency. Wrong-namespace/stale snapshot risk still needs the source-owner procedure; hashes establish byte identity, not...

Access: Read-only configured input roots; create/write/delete staging and own lock/status only inside dedicated output root; never alter inputs or existing committed...

F5 - Minimized data and retention

Choice: Memory-only snapshots; retain output CSV, hashes, count/manifest and error codes/row numbers locally for proposed 30 days. No customer names, addresses or raw rows in logs.

Rejected: Rejected cloud warehouse/vector store: no retrieval or analytics need and extra data movement.

Trade-off: Memory/row limits are provisional; real export profiling and retention approval remain launch gates. All cloud alternatives face the same access, schema, provenance...

Access: Read-only configured input roots; create/write/delete staging and own lock/status only inside dedicated output root; never alter inputs or existing committed...

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

Model cost

Illustrative model cost is USD 0.00000 per case. Unknown prices are not free service. Confirm rates, traffic, staff and tools before launch.

Full call counts, token budgets, rejected choices and costs: design.json, architecture.md and assumptions.md.

Evaluation coverage matrix

Risk / task	EV1	EV2	EV3	EV4	EV5	EV6	EV7	EV8	EV9	EV10	EV11	EV12
FM1	X	-	X	X	X	-	-	-	-	-	-	-
FM2	X	X	-	-	-	X	-	-	-	-	-	-
FM3	-	-	-	-	-	-	X	X	X	X	X	X
FM4	-	-	-	-	-	-	-	-	-	-	-	-
FM5	-	-	-	-	-	-	-	-	-	-	-	-
FM6	-	-	-	-	X	-	-	-	-	-	-	-
FM7	-	-	-	-	-	-	-	-	-	-	-	-
FM8	-	-	-	-	-	-	-	-	-	-	-	-
FM9	-	-	-	-	-	-	-	-	-	-	-	-
T2	X	-	X	X	X	-	X	X	X	X	X	X
T3	X	X	-	-	-	X	-	-	-	-	-	-
T4	X	-	-	-	X	-	-	-	-	-	-	-

Evaluation coverage matrix - continued

Risk / task	EV13	EV14	EV15	EV16	EV17	EV18	EV19	EV20	EV21	EV22	EV23	EV24
FM1	-	-	-	-	-	-	-	-	-	-	-	-
FM2	-	-	-	-	-	-	-	-	-	-	-	-
FM3	-	-	-	-	-	-	X	-	-	X	-	-
FM4	X	X	X	X	-	-	-	X	-	-	X	-
FM5	-	-	-	-	X	X	-	-	-	-	-	X
FM6	-	-	-	-	-	-	-	X	X	-	-	-
FM7	-	-	-	X	-	-	-	-	-	X	-	-
FM8	-	-	-	-	-	-	-	-	-	-	-	-
FM9	-	-	-	-	-	-	-	-	-	-	-	-
T2	-	-	-	-	-	-	X	X	X	X	-	-
T3	-	-	-	X	-	-	-	-	-	X	-	-
T4	X	X	X	X	X	X	-	X	X	X	X	X

Evaluation coverage matrix - continued

Risk / task	EV25	EV26	EV27	EV28	EV29
FM1	-	-	-	-	-
FM2	-	-	-	-	-
FM3	-	-	-	-	-
FM4	-	-	-	-	-
FM5	X	-	-	-	-
FM6	-	-	-	-	-
FM7	-	X	X	-	-
FM8	-	-	-	X	-
FM9	-	-	-	-	X
T2	-	-	-	-	-
T3	-	X	X	-	-
T4	X	X	X	-	X

X means the test is linked to that risk or task. It does not mean the test passed. Complete fixtures and assertions: eval-set.json.

Evaluation plan

Test	Layer	Purpose
EV1	golden	Positive one-row late delivery
EV2	golden	Exactly 24 hours is not late
EV3	golden	Duplicate promised ID fails even with identical rows
EV4	golden	Duplicate actual ID fails
EV5	golden	Formula-like ID is inert and rejected
EV6	golden	UTC day, sorting, leading zero and early delivery
EV7	golden	Delivered order missing promise
EV8	golden	Invalid actual date
EV9	golden	Non-UTC actual offset
EV10	golden	Missing required header
EV11	golden	Duplicate header
EV12	golden	CSV row width mismatch
EV13	golden	Known pre-publication disk-full
EV14	golden	Unknown commit reconciles without duplicate publish
EV15	golden	Proven absent permits one retry
EV16	golden	Unresolved publication is blocked
EV17	golden	Concurrent run cannot write
EV18	golden	Changed input never overwrites report
EV19	golden	Undelivered and promised-only produce empty valid CSV
EV20	golden	Revoked local write permission fails safely
EV21	golden	Symlink path cannot escape input scope
EV22	golden	Input size exceeds configured bound
EV23	golden	Retry budget cannot loop
EV24	golden	Exact rerun returns durable no-op
EV25	golden	Stale lock is never auto-broken
EV26	golden	Cancellation before publication
EV27	golden	Cancellation after commit start defers to proof
EV28	golden	Local status failure remains visible
EV29	golden	Extra data is minimized from outputs and logs
EV30	offline	Offline parser, invariants and end-to-end suite
EV31	online	Daily local report and failure monitor
EV32	release	Release gate and deployment filesystem proof

These are test plans. No business controls, provider quality or user outcomes were measured. See eval-set.json for fixtures and pass rules.

Measurement and rollout

R1 - Exact report accuracy and business evidence

Measure: Daily record accuracy = correctly included/flagged rows divided by independently expected rows; track missing and extra rows separately so denominator cannot hide omissions. Weekly aggregate is secondary to any integrity...

Signals: No AI metrics. Exact join/UTC fixture pass fraction causally protects correct queue selection by eliminating date/identity errors. Source freshness is separately required: correct arithmetic...

R2 - Completeness and durable report KPI

Measure: Publication completeness = expected report dates with one checksum-verified COMPLETE directory by 10:00 UTC / expected scheduled report dates in each UTC week; blocked/cancelled/missing dates remain in denominator. Baseline unknown...

Signals: Guardrail: false-complete count must be 0; correct durable proof makes a report safely consumable. Busy/conflict/unknown counts diagnose reliability, not delivery performance.

R3 - Manual review effort

Measure: Operations review effort = minutes from export pair availability to verified daily queue, including correction/investigation time, per UTC report day. Baseline unknown; collect manual timers during shadow. Target hypothesis: reduce...

Signals: Runtime p95 under provisional 120s and output verification time are operational metrics. Runtime affects availability but is not a substitute for correctness. Human ack...

R4 - Shadow then assist-only rollout

Phase / owner: Shadow → Assist-only → Routine assist; no autonomous operations action phase. / Operations report owner owns stop/restore; Implementation owner repairs; Data owner...

Gate: Shadow entry: implemented CLI/harness and EV32 local synthetic/dependency evidence complete; no user consumption. Shadow exit: five consecutive complete daily export pairs spanning approved source strata, exact manually adjudicated report agreement and no...

R5 - Measure effort with paired days

Test: Unit is UTC report date; no randomization of customer service. Operators must not share report outputs before timing. If independent operators unavailable, record limitation and run descriptive timings without causal savings claim.

Decision gate: Eligibility: complete approved source pairs in the requested UTC day; strata by row-count bands and late-order prevalence measured from sources. Each day...

R6 - Accountability and manual recovery

Phase / owner: All phases / Proposed roles only; appointment and capacity require H5/H7/H8 evidence.

Gate: Before launch each proposed role must be assigned to an actual person. Report owner checks by 10:00 UTC, ack by 12:00 UTC and manual escalation to operations lead thereafter. No automated messaging...

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

Build handoff and review

Start with build/00-overview.md. Load one milestone file at a time. Exact source rules remain in design.json; do not treat design approval as live write permission.

X1: Implement only the final agreed design, preserving IDs in code docs/test names. This package is proposed design, not the report implementation or an executed operational run.

Review dimensions - evidence in review.json

Dimension	Score	Dimension	Score
D1	3/4	D7	3/4
D2	3/4	D8	4/4
D3	3/4	D9	3/4
D4	3/4	D10	3/4
D5	3/4	D11	3/4
D6	3/4	D12	3/4

ISSUE1 major resolved: Actual helper validate and same-folder export both exited 0; round-1 evidence is saved. Semantic payload review then found two material implementation ambiguities below.

ISSUE2 major resolved: CTX4/C1/C4 did not give one exact typed manifest object or exact config-fingerprint key set, yet C4 delegated the manifest to CTX4. Different implementers could...

ISSUE3 major resolved: C4 reused one staging path across its allowed retry after proven absence, so an orphan first-attempt staging directory could block EV15 despite the designed...

Glossary

canonical: agreed

discriminated: typed

payload binding: approved details

digest: fingerprint

idempotency: duplicate-write protection

reciprocal: two-way

terminal disposition: final outcome

calc_digest: a fingerprint of the approved amount

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

Full assumptions, owners and evidence: assumptions.md. Review method and failed rounds are preserved in review.json and review-round-*.json.